Assignment 5

Problems:

1. Let V be a vector space over F. Show that the sum of two inner products on V is an inner product on V. Is the difference of two inner products an inner product? Show that a positive multiple of an inner product is an inner product.

2. Let V be a real or complex vector space with an inner product. Show that the quadratic form determined by the inner product satisfies the *parallelogram law*

$$\|\alpha+\beta\|^2+\|\alpha-\beta\|^2=2\|\alpha\|^2+2\|\beta\|^2.$$

3. Show that the formula

$$\left\langle \sum_j a_j x^j \,\middle|\, \sum_k b_k x^k \right\rangle = \sum_{j,k} \frac{a_j b_k}{j+k+1} \tag{1}$$

defines an inner product on the space $R[x]$ of polynomials with real coefficients. Let W be the subspace of polynomials of degree less than or equal to n. Find the weight matrix associated with this inner product on W, relative to the ordered basis $1, x, x^2, \ldots, x^n$. (*Hint:* To show that the formula defines an inner product, observe that

$$\langle f|g\rangle = \int_0^1 f(t)g(t)dt \tag{2}$$

and work with the integral.)

4. Let V be the vector space of real polynomials with degree at most 3. Equip V with the inner product

$$\langle f|g\rangle = \int_0^1 f(t)g(t)dt.$$

   (a) Find the orthogonal complement of the subspace of constant polynomials.

   (b) Apply the Gram-Schmidt process to the basis $\{1, x, x^2, x^3\}$.

5. Let V be the vector space of all $n\times n$ matrices over C, with the inner product $(A|B) = \operatorname{tr}(AB^H)$. Find the orthogonal complement of the subspace of diagonal matrices.

6. Let W be a finite-dimensional subspace of an inner product space V, and let E be the orthogonal projection of V in W. Prove that $\langle E\alpha|\beta\rangle = \langle \alpha|E\beta\rangle$ for all α, β in V.